

API Specification Doc

“Fábula Dental” Management System

Version	Date	Author	Description
1.0	02-Jun-2026	CodeXplorers Team	URIs list

CRUD server IP address: 34.28.34.201

CRUD server port: 3000

Business Logic server IP address: 34.122.231.144

Business Logic server port: 3001

Methods

Business Logic

<http://34.122.231.144:3001>

Patients

1. Calculate patient pediatric category

Takes a patient's date of birth as an input, calculates the exact age in years and months, and automatically maps the patient to their respective clinical age group classification for safe dosage validation.

Request

Method	URL
POST	/fabuladental/patients/calculate-category

Response

HTTP code	Response
200	{"calculatedAgeYears": 5, "calculatedAgeMonths": 3, "pediatricCategory": "Pre-schooler", "dosageRestrictionFactor": "0.5x"}
400	{"error": "Valid birthday format required."}
500	{"error": "Internal processing error calculating clinical category."}

Payments

2. Get payment history

Retrieves the payment history and calculates current status (Pending, Partial, Completed).

Request

Method	URL
GET	/fabuladental/payments

Response

Http code	Response
200	[{"paymentID": "...", "amount": 50.0, "status": "Partial"}]
500	{"error": "Unable to fetch payment history."}

3. Retrieve a specific payment record

Retrieves the complete details of a particular payment record by its ID, returning the data as a JSON object.

Request

Method	URL
GET	fabuladental/payments /:paymentId

Response

Http Code	Response
200	{"success": true, "data": {"id": "p_98765", "patientId": "1727095414", "amount": 150.00, "type": "Deposit", "status": "Pending", "date": "2023-10-27"}}

400	<code>{"error": "Payment ID required."}</code>
404	<code>{"error": "Payment not found."}</code>
500	<code>{"error": "Failed to retrieve payment details."}</code>

4. Retrieve all payment records for a specific patient

Query and return the complete history of all payment records associated with a specific patient using their ID. The response is delivered as a JSON object containing a list.

Request

Method	URL
GET	<code>fabuladental/patients /:paymentId/payments</code>

Response

Http Code	Response
200	<code>{"success": true, "data": [{"id": "p_98765", "amount": 150.00, "type": "Deposit", "status": "Pending", "date": "2023-10-27"}, {"id": "p_98766", "amount": 300.00, "type": "Final Payment", "status": "Completed", "date": "2023-11-15"}]}</code>
400	<code>{"error": "Invalid Patient ID parameter."}</code>
404	<code>{"error": "Patient not found or no payment records discovered."}</code>
500	<code>{"error": "Failed to retrieve patient payment history."}</code>

Supplies

5. Update supplies expiration status

Compares the `expirationDate` of all inventory items against the current system date, automatically calculating and updating the `status` field (`Current`, `NextExpiration`, `Expired`) in the database.

Request

Method	URL
POST	<code>/fabuladental/supplies/status-calculations</code>

Response

HTTP code	Response
200	<code>{"success": true, "message": "Supplies expiration status successfully updated.", "processedItems": 45}</code>
500	<code>{"error": "Failed to calculate or update expiration statuses."}</code>

6. Get inventory total asset valuation

Calculates the total monetary value of the active warehouse stock in real time by multiplying each item's current `quantity` by its registered `unitCost`.

Request

Method	URL
GET	<code>/fabuladental/supplies/asset-valuations</code>

Response

HTTP code	Response
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200	<code>{"totalInventoryValue": 3420.75, "currency": "USD", "calculatedAt": "2026-06-09T19:30:00Z"}</code>
500	<code>{"error": "Unable to compile asset valuation data. "}</code>

7. Get restock provisioning and budget analysis

Evaluates all items with a quantity less than or equal to 5, calculates the units needed to reach the clinic's ideal stock of 15 units per item, and sums their estimated costs to deliver a consolidated replenishment budget.

Request

Method	URL
GET	<code>/fabuladental/supplies/restock-provisions</code>

Response

HTTP code	Response
200	<code>{"consolidatedRestockBudget": 415.50, "currency": "USD", "itemsToPurchase": [{"id": 4, "supplyName": "Anesthesia", "unitsToOrder": 12, "estimatedCost": 264.00}, {"id": 15, "supplyName": "Resin Z350", "unitsToOrder": 10, "estimatedCost": 151.50}]}</code>
500	<code>{"error": "Unable to calculate restock provisioning analysis."}</code>

8. Calculate expired supplies financial loss

Calculates the total direct financial loss of the clinic by summing the total cost (quantity * unitCost) of all inventory items whose status is currently marked as 'Expired'.

Request

Method	URL
GET	<code>/fabuladental/supplies/expiration-losses</code>

Response

HTTP code	Response
200	<pre>{"totalLossAmount": 185.00, "currency": "USD", "expiredItemsCount": 2, "calculatedAt": "2026-06-09T19:30:00Z"}</pre>
500	<pre>{"error": "Could not calculate historical expiration losses."}</pre>

9. Get next-expiration capital at risk assessment

Quantifies the total financial value at risk of being wasted by multiplying the quantity by the unitCost of all items whose current status is marked as 'NextExpiration'.

Request

Method	URL
GET	/fabuladental/supplies/capital-risks

Response

HTTP code	Response
200	<pre>{"totalCapitalAtRisk": 240.00, "currency": "USD", "riskPeriod": "30 Days or Less", "itemsNearExpirationCount": 3}</pre>
500	<pre>{"error": "Failed to compile the expiration capital risk assessment."}</pre>

CRUD Operations

<http://34.28.34.201:3000>

Payments

1. Update payment history

Updates a recorded payment to correct errors or update status.

Request

Method	URL
PUT	/fabuladental/payments/:paymentId

Response

Http Code	Response
200	{"success": true, "message": "Payment updated"}
400	{"error": "Invalid payment ID or data."}
404	{"error": "Payment record not found."}

2. Delete a payment record

Deletes a payment record from history.

Request

Method	URL
DELETE	/fabuladental/payments

Response

Http Code	Response
200	<code>{"success": true, "message": "Payment deleted"}</code>
400	<code>{"error": "Payment ID required."}</code>
404	<code>{"error": "Payment not found."}</code>
500	<code>{"error": "Deletion failed."}</code>

3. Record a payment

Records a new patient payment (Deposit or Final).

Request

Method	URL
POST	<code>/fabuladental/payments</code>

Response

Http code	Response
201	<code>{"success": true, "message": "Payment recorded"}</code>
400	<code>{"error": "Invalid payment data."}</code>
500	<code>{"error": "Could not record payment."}</code>

Patients

4. Register a new patient

Registers a new patient with ID, name, birthday, and reason for consultation.

Request

Method	URL
POST	/fabuladental/patients

```
{  
  "patientID": "1727095414",  
  "fullName": "Victoria Diaz",  
  "phone": "0984284522",  
  "email": "cvdiaz3@espe.edu.ec",  
  "birthday": "20/03/2003",  
  "gender": "Female",  
  "reasonForConsultation": "gum pain"  
}
```

Response

HTTP code	Response
201	{"success": true, "message": "Patient registered successfully"}
400	{"error": "Missing required fields."}
500	{"error": "Something went wrong. Please try again later."}
201	{"success": true, "message": "Patient registered successfully"}

5. Get all patients

Retrieves the list of all registered patients to populate the dentist's table dynamically.

Request

Method	URL
GET	/fabuladental/patients

Response

HTTP code	Response
200	[{"patientID": "1727095414", "fullName": "Victoria Diaz", "phone": "0984284522", "email": "cvdiaz3@espe.edu.ec", "birthday": "20/03/2003", "gender": "Female", "reasonForConsultation": "gum pain"}, {"patientID": "1700799198", "fullName": "Juan Perez", "phone": "098434532", "email": "jperz@gmail.com", "birthday": "02/04/2013", "gender": "Male", "reasonForConsultation": "caries"}]
400	{"error": "Unable to fetch patients."}

6. Get patient by ID

Retrieves the full information of a single patient identified by patientID.

Request

Method	URL
GET	/fabuladental/patients/:patientID

```
{"patientID": "1727095414"}
```

Response

HTTP code	Response
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200	<code>{"patientID": "1727095414", "fullName": "Victoria Diaz", "phone": "0984284522", "email": "cvdiaz3@espe.edu.ec", "birthday": "20/03/2003", "gender": "Female", "reasonForConsultation": "gum pain"}</code>
404	<code>{"error": "Patient not found."}</code>
505	<code>{"error": "Unable to fetch patient."}</code>

7. Update patient information

Updates an existing patient's information directly from the table.

Request

Method	URL
PUT	<code>/fabuladental/patients/:patientId</code>

```
{
  "patientID": "1727095414",
  "fullName": "Victoria Diaz",
  "phone": "0984284522",
  "email": "cvdiaz3@espe.edu.ec",
  "birthday": "20/03/2003",
  "gender": "Female",
  "reasonForConsultation": "gum pain"
}
```

Response

HTTP code	Response
200	<code>{"success": true, "message": "Patient updated"}</code>
400	<code>{"error": "Patient ID is required."}</code>
404	<code>{"error": "Patient not found."}</code>

500	{"error": "Something went wrong."}
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8. Delete a patient

Deletes a patient record.

Request

Method	URL
DELETE	/fabuladental/patients/:patientId

Type	Params	Values
URL_PARAM	id	number

Response

HTTP code	Response
200	{"success": true, "message": "Patient deleted"}
400	{"error": "Invalid or missing patient ID."}
404	{"error": "Patient not found."}
500	{"error": "Deletion failed."}

Supplies

9. Get supplies list

Retrieves the complete list of dental supplies directly from the database, returning all persisted fields matching the database schema.

Request

Method	URL
GET	/fabuladental/supplies

Response

HTTP code	Response
200	[{"id": 1, "supplyName": "Resina", "quantity": 24, "unitCost": 3.75, "orderDate": "2026-05-10", "expirationDate": "2027-05-10", "status": "Expired"}, ...]
500	{"error": "Unable to fetch inventory."}

10. Get supplies filtered by maximum quantity

A CRUD query that reads and filters supplies from the database whose current stock is less than or equal to the maximum quantity specified in the URL path.

Request

Method	URL
GET	/fabuladental/supplies/quantity-thresholds/:maxQuantity

Response

HTTP code	Response
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200	[{"id": 1, "supplyName": "Resina", "quantity": 3, "unitCost": 3.75, "orderDate": "2026-05-10", "expirationDate": "2027-05-10", "status": "Expired"}, ...]
400	{"error": "Invalid maximum quantity format."}
500	{"error": "Database error while filtering supply quantities."}

11. Get supplies filtered by database status

A CRUD query that retrieves records from the database matching the exact status string passed in the URL path (`Expired` or `NextExpiration`)

Request

Method	URL
GET	/fabuladental/supplies/statuses/:statusValue

Response

HTTP code	Response
200	[{"id": 1, "supplyName": "Resina", "quantity": 24, "unitCost": 3.75, "orderDate": "2026-05-10", "expirationDate": "2027-05-10", "status": "Expired"}, ...]
400	{"error": "Provided status value does not exist."}
500	{"error": "Database error while filtering supply statuses."}

12. Register a dental supply

Registers a new dental supply into the inventory with quantity and dates.

Request

Method	URL
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POST	/fabuladental/supply
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```
{
  "supplyName": "Resin",
  "quantity": 10,
  "unitCost": 5.50,
  "orderDate": "2026-06-09",
  "expirationDate": "2027-06-09"
}
```

Response

HTTP code	Response
201	{"success": true, "message": "Supply added"}
400	{"error": "Missing supply data."}
500	{"error": "Could not add supply."}

13. Update supply

Edits an existing supply in the inventory by updating any of its modifiable attributes.

Request

Method	URL
PUT	/fabuladental/supplies/:id


```
{  
  
  "supplyName": "Resin Z350",  
  
  "quantity": 15,  
  
  "unitCost": 6.25,  
  
  "orderDate": "2026-06-09",  
  
  "expirationDate": "2027-06-09"  
}
```

Response

HTTP code	Response
200	{"success": true, "message": "Supply updated"}
400	{"error": "Invalid supply ID or quantity."}
404	{"error": "Supply not found."}

14. Delete a dental supply

Removes a dental supply from the inventory using its unique ID.

Request

Method	URL
DELETE	/fabuladental/supplies/:id

Response

HTTP code	Response
200	{"success": true, "message": "Supply deleted"}

400	<code>{"error": "Supply ID required."}</code>
404	<code>{"error": "Supply not found."}</code>
500	<code>{"error": "Deletion failed."}</code>

Glossary

Conventions

- **Client** - Client application.
- **Status** - HTTP status code of response.
- All responses are in JSON format except where HTML views are returned.
- All request parameters are mandatory unless marked as [optional].
- All request parameters are mandatory unless explicitly marked as [optional]
- The API includes both RESTful endpoints (data operations) and HTML views (forms and dashboards).
- Authentication is required for most endpoints except the public treatments page and login page.

Status Codes

All status codes are standard HTTP status codes. The below ones are used in this API.

2XX - Success of some kind

4XX - Error occurred in client's part

5XX - Error occurred in server's part

Status Code	Description
200	OK
201	Created
400	Bad request
401	Authentication failure
403	Forbidden
404	Resource not found
500	Internal Server Error